🛡️ Equivalence Class Partitioning (ECP) — Automated Selenium Suite

Full-Stack Verification of Input Domains, Validation Boundaries, and AI Triage Heuristics

TOTAL PARTITIONS	PASSED	FAILED	SUCCESS RATE
17	17	0	100.0%

TEST ID	SUBSYSTEM	INPUT DOMAIN	PARTITION TYPE	TEST VALUES	EXPECTED OUTCOME	STATUS	SCREENSHOT
ECP-REG-01	Civilian Registration	Full Name, Email, Password, Confirm	Valid (VEC)	Name: 'Maria Santos', Email: 'unique@safety.org', Pwd: 'Passcode123!'	Form submits successfully, civilian account created, dashboard reached	PASSED	View PNG
ECP-REG-02	Civilian Registration	Full Name	Invalid (IEC: Length < 2)	Name: 'X' (1 character)	Validation error: 'Name must be at least 2 characters'	PASSED	View PNG
ECP-REG-03	Civilian Registration	Email Address	Invalid (IEC: Non-RFC email)	Email: 'invalid-email-address' (missing @ and domain)	HTML5 constraint & React error: 'Valid email is required'	PASSED	View PNG
ECP-REG-04	Civilian Registration	Password	Invalid (IEC: Length < 8)	Password: '12345' (5 characters)	Validation error: 'Password must be at least 8 characters'	PASSED	View PNG
ECP-REG-05	Civilian Registration	Confirm Password	Invalid (IEC: Mismatched values)	Password: 'CorrectPass123!', Confirm: 'DifferentPass456!'	Validation error: 'Passwords do not match'	PASSED	View PNG
ECP-AUTH-01	Authentication / Login	Civilian Credentials	Valid (VEC)	Email: 'civilian@demo.com', Password: 'demo1234'	Authentication succeeds, redirected to '/civilian/dashboard'	PASSED	View PNG
ECP-AUTH-02	Authentication / Login	Commander Credentials	Valid (VEC: Role RBAC)	Email: 'commander@demo.com', Password: 'demo1234'	Authentication succeeds, redirected to '/commander/command-center'	PASSED	View PNG
ECP-AUTH-03	Authentication / Login	Password	Invalid (IEC: Incorrect Password)	Email: 'civilian@demo.com', Password: 'WrongSecret999!'	Access denied, alert banner displays 'Login failed'	PASSED	View PNG
ECP-AUTH-04	Authentication / Login	Email Address	Invalid (IEC: Unregistered Account)	Email: 'unknown_ghost_account@nonexistent.org'	Access denied, alert banner displays 'Login failed'	PASSED	View PNG
ECP-AUTH-05	Authentication / Login	Email & Password Fields	Invalid (IEC: Empty Inputs)	Email: '', Password: ''	Client errors: 'Email is required' & 'Password is required'	PASSED	View PNG
ECP-INC-01	Incident Intake & AI Triage	Description	Invalid (IEC: Length < 5)	Description: 'Fire' (4 characters)	Validation error: 'Description must be at least 5 characters'	PASSED	View PNG
ECP-INC-02	Incident Intake & AI Triage	Coordinates	Invalid (IEC: Empty Coordinates)	Latitude: '', Longitude: ''	Errors: 'Valid latitude is required' & 'Valid longitude is required'	PASSED	View PNG

TEST ID	SUBSYSTEM	INPUT DOMAIN	PARTITION TYPE	TEST VALUES	EXPECTED OUTCOME	STATUS	SCREENSHOT
ECP-INC-03	Incident Intake & AI Triage	AI Severity Scoring (Critical)	`Valid (VEC: Critical Class 7-10)`	`Building Collapse, Trapped victims, Fire, Casualties: 3, Injuries: 12`	Severity 7-10/10, Priority 'High', AI Recommendation computed	PASSED	View PNG
ECP-INC-04	Incident Intake & AI Triage	AI Severity Scoring (Low)	`Valid (VEC: Low Class 1-5)`	`Minor Flood, puddle on curb, Casualties: 0, Injuries: 0, No critical tags`	Severity 1-5/10, Priority 'Low', Incident registered	PASSED	View PNG
ECP-RES-01	Resource Allocation	Quantity	`Valid (VEC: 1 <= Qty <= Stock)`	`Quantity: 2 (within available inventory)`	Confirm Allocation button enabled, remaining stock projected	PASSED	View PNG
ECP-RES-02	Resource Allocation	Quantity	`Invalid (IEC: Quantity <= 0)`	`Quantity: 0`	Confirm Allocation button disabled by business boundary logic	PASSED	View PNG
ECP-RES-03	Resource Allocation	Quantity	`Invalid (IEC: Over-Allocation)`	`Quantity: 999999 (exceeds inventory)`	Confirm button disabled, warning: 'exceeds available inventory'	PASSED	View PNG